Further, it may be noted that the odds are a : b in favour of an event is the same as to say that the odds are b : a against the event.

If the probability of an event is p , then the odds in favour of its occurrence are p to $(1-p)$ and the odds against its occurrence are $(1-p)$ to p .

Illustration 12.7

(i) Suppose a die is rolled.

Let S be the sample space and A be the event of getting 5.

$n(S)=6$, $n(A)=1$ and $n(\bar{A})=5$.

It can also be interpreted as

Odds in favour of A is $1:5$ or $\frac{1}{5}$, odds against A is $5:1$ or $\frac{5}{1}$,

and $P(A)=\frac{n(A)}{n(A)+n(\bar{A})}=\frac{1}{5+1}=\frac{1}{6}=\frac{n(A)}{n(S)}$.

(ii) Suppose B is an event such that odds in **favour** of B is 3:5, then $P(B)=\frac{3}{8}$

(iii) Suppose C is an event such that odds **against** C is 4:11, then $P(C)=\frac{11}{15}$.

Example 12.11

A man has 2 ten rupee notes, 4 hundred rupee notes and 6 five hundred rupee notes in his pocket. If 2 notes are taken at random, what are the odds in favour of both notes being of hundred rupee denomination and also its probability?

Solution

Let S be the sample space and A be the event of taking 2 hundred rupee note.

Therefore, $n(S)=12c_2=66$, $n(A)=4c_2=6$ and $n(\bar{A})=66-6=60$

Therefore, odds in favour of A is 6: 60

That is, odds in favour of A is 1: 10, and $P(A)=\frac{1}{11}$.

EXERCISE 12.1

(1) An experiment has the four possible mutually exclusive and exhaustive outcomes A, B, C, and D. Check whether the following assignments of probability are permissible.

(i) $P(A)$ = 0.15, $P(B)$ = 0.30, $P(C)$ = 0.43, $P(D)$= 0.12

(ii) $P(A)$ = 0.22, $P(B)$ = 0.38, $P(C)$ = 0. 16, $P(D)$ = 0.34

(iii) $P(A)=\frac{2}{5}$, $P(B)=\frac{3}{5}$, $P(C)=-\frac{1}{5}$, $P(D)=\frac{1}{5}$

(2) If two coins are tossed simultaneously, then find the probability of getting

(i) one head and one tail (ii) atmost two tails

(3) Five mangoes and 4 apples are in a box. If two fruits are chosen at random, find the probability that (i) one is a mango and the other is an apple (ii) both are of the same variety.

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